Cross-Space Distillation (Bridge)

Mapping arXiv:2606.32020 onto XOPD's Cross-VAE Inverse Problem

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Abstract

Companion note to `cross_latent_distillation_problem.tex`, `avoid_inverse_transport_analysis.tex`, and `inverse_mapping_methods.tex`. We summarize Nguyen et al., Cross-Space Distillation (arXiv:2606.32020) — a method that trains a lightweight **Bridge** $\mathcal{B}_\phi : \mathcal{Z}_S \to \mathcal{Z}_T$ so that compact one-step Students (different resolution / VAE) can run standard distribution-based distillation (VSD / GAN, DMD2-style) against modern Teachers. Figures are taken from the arXiv e-print source (`assets/`). The key mapping for our codebase: Bridge is a concrete construction of the *hard* student→teacher inverse Q (pixel-anchored spatial prior + learned projector + attention fidelity), but it is designed for *measure / score matching in Teacher space*, not for XOPD's pointwise velocity pushforward L1. That places it closest to Route A in the avoid-inverse note, not to linear-P L1 transport.

1 Problem they formalize

Distribution-based one-step distillation (VSD / adversarial / DMD-family) almost always assumes a **Shared-Space** constraint: Teacher and Student share the same VAE and latent grid, so Teacher scores / discriminators can be evaluated on Student samples directly. Modern Teachers (SD 3.5, FLUX.2, ...at 1024^2) and compact Students (SD 1.5 / 2.1 at 512^2) violate this in two coupled ways:

1. **Cross-Resolution:** $z_S \in \mathbb{R}^{h\times w\times C_S}$ vs. $z_T \in \mathbb{R}^{H\times W\times C_T}$ with $(h, w) \neq (H, W)$.
2. **Cross-VAE:** different autoencoders $\Rightarrow \mathcal{Z}_S \ncong \mathcal{Z}_T$ (channels, statistics, semantics).

They call the joint regime **Cross-Space Distillation**. Empirically they argue that cross-architecture (UNet vs. DiT) and cross-mechanism (DDPM vs. flow matching) are secondary once outputs are reparameterized to a common x_0 estimate in a shared (Teacher) latent; the *blocking* mismatch is resolution + VAE.

2 Bridge construction

Object. A frozen (after training) map

$$\hat{z}_T \;=\; \mathcal{B}_\phi(z_S) \;\in\; \mathcal{Z}_T, \tag{1}$$

so Student outputs become Teacher-compatible.

Architecture (spatial prior + projector). Rather than learning a latent upsampler from scratch, they freeze the first n blocks of the Student decoder $\mathcal{D}_S^{(n)}$ (experiments: n=1) as a *spatial prior* that expands the Student grid to the Teacher resolution, then train only a compact projector g_ϕ (SwinIR, ∼5M params):

$$f_{\text{prior}} = \mathcal{D}_S^{(n)}(z_S) \in \mathbb{R}^{H\times W\times C_{\text{prior}}}, \tag{2}$$

$$\hat{z}_T = \mathcal{B}_\phi(z_S) = g_\phi(f_{\text{prior}}) \in \mathbb{R}^{H\times W\times C_T}. \tag{3}$$

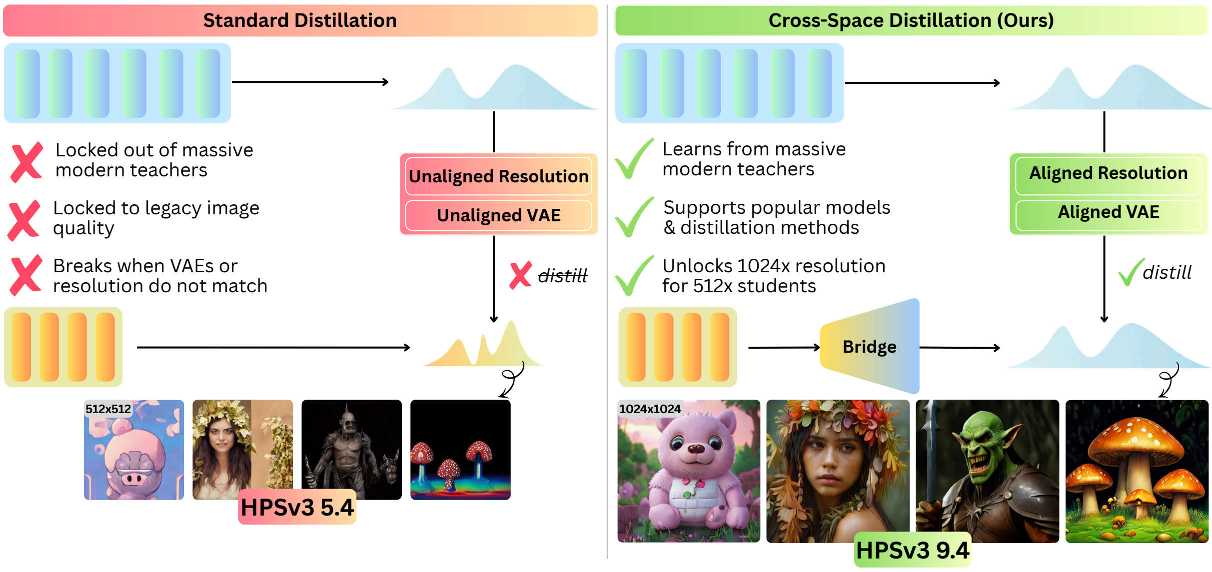

Figure 1: Shared-Space vs. Cross-Space. Existing distribution distillation assumes a shared latent; Bridge $\mathcal{B}_\phi$ maps $z_S \mapsto \hat{z}_T$ so standard one-step objectives apply without changing the Student backbone. (Figure from arXiv:2606.32020 source `assets/teaser.pdf`.)

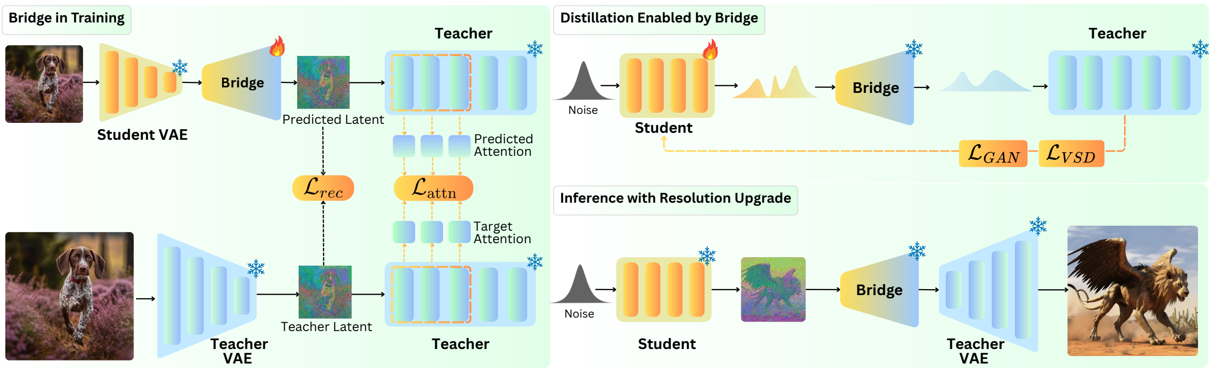

Figure 2: Left: Bridge training with $\mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{attn}}$. Top-right: freeze Bridge, run VSD/GAN in Teacher space. Bottom-right: optional inference upgrade — decode $\hat{z}_T$ with Teacher decoder at Teacher resolution. (Figure from arXiv:2606.32020 source `assets/system.pdf`.)

Pipeline: $z_S \xrightarrow{\mathcal{D}_S^{(n)}} f_{\text{prior}} \xrightarrow{g_\phi} \hat{z}_T$.

Training objectives (offline, paired images). Encode the same image with both VAEs to get (z_S, z_T). Minimize

$$\mathcal{L}_{\text{rec}} = \|z_T - \hat{z}_T\|_1 , \tag{4}$$

$$\mathcal{L}_{\text{attn}} = \sum_{l \in \text{Layers}} \tau^2 \, \text{KL}\big(P^l(\hat{z}_T; t, c) \,\big\|\, P^l(z_T; t, c)\big), \tag{5}$$

$$\mathcal{L}_{\text{final}} = \alpha \, \mathcal{L}_{\text{rec}} + \beta \, \mathcal{L}_{\text{attn}}, \tag{6}$$

with $\alpha{=}\beta{=}1$, attention temperature $\tau{=}3$, and noising level $t{=}1$ for attention. P^l are Teacher self-attention row-softmax maps; reverse KL follows MiniLLM (emphasize dominant mass). Ablations: ℓ_1 alone is over-smooth; E-LatentLPIPS helps little; Attention Fidelity restores fine structure (eyes, fur, edges).

Usage after Bridge is frozen.

- **Distillation:** Student generates z_S; map $\hat{z}_T = \mathcal{B}_\phi(z_S)$; apply DMD2-style $\mathcal{L}_{\text{VSD}} + \mathcal{L}_{\text{GAN}}$ in

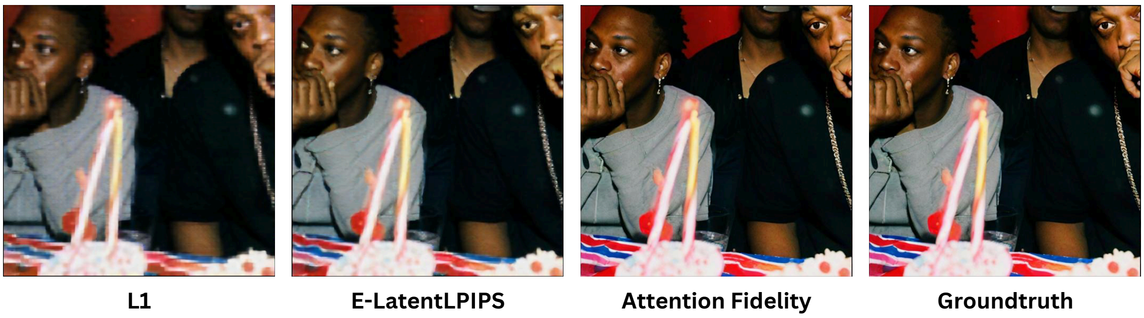

Figure 3: Objective ablation (SD 2.1→SDXL Bridge): ℓ_1 only / +E-LatentLPIPS / +Attention Fidelity vs. GT decode. (Figure from arXiv:2606.32020 source `assets/loss.pdf`.)

Teacher space (Teacher score / discriminator see $\hat{z}_T$).

- **Inference upgrade (optional):** same Bridge + Teacher decoder yields 1024^2 outputs from a 512^2 Student without changing the denoiser.

3 Reported evidence (for our design priors)

Students: one-step SD 1.5 (DMD2 init) and SD 2.1 (SiD). Teachers: SDXL, Kolors, PixArt-Σ, FLUX.2-klein-4B, SD 3.5 M (all 1024^2). Bridge trained on ∼2M Teacher-synthesized images (JourneyDB/LAION prompts). Headline: SD 1.5 HPSv3 5.37 → 9.42 (SD 3.5 M teacher), merged average of five distilled students reaches 10.53. Mechanism-agnostic x_0 reparameterization enables flow Teachers → DDPM Students without layer-wise matching.

4 Map onto XOPD notation

Recall our notation (`cross_latent_distillation_problem.tex`): forward $\mathsf{T} : \mathcal{Z}_T \to \mathcal{Z}_S$ (teacher→student, easy when $d_T > d_S$), inverse $\mathsf{T}^{-1} : \mathcal{Z}_S \to \mathcal{Z}_T$ (hard). Ideal on-manifold lift $\bar{z}_T(z_S) = \mathcal{E}_T(\mathcal{D}_S(z_S))$.

Observation 1 (Bridge is a learned inverse Q, not a forward T)**.** $\mathcal{B}_\phi : \mathcal{Z}_S \to \mathcal{Z}_T$ is exactly the student→teacher direction we call Q / T^{-1}. The frozen $\mathcal{D}_S^{(n)}$ is an architectural prior for *spatial* expansion; g_ϕ does channel/semantic alignment. Training pairs are $(z_S, z_T) = (\mathcal{E}_S(x), \mathcal{E}_T(x))$, so the regression target is the **pixel-anchored** Teacher encode — the same ideal lift $\bar{z}_T$ that our notes treat as the on-manifold gold standard — plus an attention regularizer in Teacher space.

Remark 1 (Why this is Route A-adjacent, not L1 pushforward)**.** `avoid_inverse_transport_analysis.tex` Route A replaces pointwise velocity targets by score / DMD-style measure matching so that supervision lives where the Teacher is defined — after a push of Student samples into Teacher space (or a shared score space). Bridge is precisely the missing interface that makes Route A *implementable* under Cross-VAE: freeze $Q = \mathcal{B}_\phi$, then run DMD2 in $\mathcal{Z}_T$. It does **not** construct a differentiable forward T with Jacobian for $(\mathsf{T}\#v_T)$ in Student coordinates; therefore it does not by itself make XOPD L1 (pointwise transition matching in $\mathcal{Z}_S$) well-posed.

Remark 2 (Relation to methods already in our inverse note)**.**

- **Pixel bridge.** For $\mathcal{E}_T(\mathcal{D}_S(z_S))$, Bridge keeps a frozen decoder *prefix* (cheaper than full decode) and replaces the Teacher encoder with a learned g_ϕ supervised with ℓ_1 + attention — a trainable refinement of the same on-manifold idea.
- vs. **MSE / linear** Q: paper confirms ℓ_2/mean-seeking is weak; they use ℓ_1 and add Teacher-attention reverse KL instead of a latent discriminator (Method 1) or cINN (Method 2).

	Cross-Space Bridge	**XOPD cross-VAE (ours)**
Primary mismatch	Cross-Res. + Cross-VAE	Cross-VAE (often also Res.)
Hard map learned	$Q = \mathcal{B}_\phi : \mathcal{Z}_S \to \mathcal{Z}_T$	Q (HSCT / disc. / cINN / ...)
Easy map	unused during distill (Student stays in $\mathcal{Z}_S$)	linear $P : \mathcal{Z}_T \to \mathcal{Z}_S$ for L0 / pushforward
Distill objective	VSD / GAN on $\hat{z}_T$ (measure matching)	L0 sample transport + L1 transition / velocity in $\mathcal{Z}_S$
Where Teacher is queried	Teacher space after Bridge	either Teacher space via Q, or Student space via $P\#$
Spatial prior	frozen $\mathcal{D}_S^{(n)}$	optional; often pixel bridge $\mathcal{E}_T(\mathcal{D}_S(\cdot))$
Extra supervision	reverse-KL on Teacher attention	cycle / adv / NLL / hidden-state cond. (HSCT)

Table 1: Bridge vs. XOPD cross-VAE: same hard inverse direction, different distill geometry.

- vs. **HSCT / M8**: HSCT conditions Q on student DiT hidden states for on-policy L1 queries; Bridge is offline, prompt/image-paired, and *does not* use student transformer features — it relies on VAE spatial prior + Teacher attention.
- vs. **same-VAE DMD** (`dmd_cross_model_design.md`): that RFC assumes shared VAE. Under Cross-VAE, Bridge (or equivalent Q) is a prerequisite before real/fake scores can share a latent.

Takeaway for Flow-Factory XOPD

If the goal is **cross-VAE one-step / DMD-style** distillation into a compact backbone: Bridge is a strong external recipe for learning Q (spatial prior + ℓ_1 + Attention Fidelity) and then running Teacher-space VSD/GAN — align with Route A, not with linear-P L1. If the goal remains **multi-step on-policy OPD with L0/L1 in Student space**: Bridge alone is insufficient; still need forward P (or pixel adjoint Route B) for Student-space losses. Attention Fidelity is a reusable objective to add on top of any learned Q (disc. / HSCT / cINN) to keep $\hat{z}_T$ Teacher-attention-compatible.

Cross References

- Paper: arXiv:2606.32020 (Cross-Space Distillation).
- Problem formalization: `docs/xopd/cross_vae/cross_latent_distillation_problem.tex`.
- Avoiding the hard inverse / Route A: `docs/xopd/cross_vae/avoid_inverse_transport_analysis.tex`.
- Inverse algorithms (disc. / cINN): `docs/xopd/cross_vae/inverse_mapping_methods.tex`.
- Method zoo (M1–M7): `docs/xopd/cross_vae/xopd_vae_space_align.tex`.
- Same-VAE DMD design: `docs/xopd/xdmd/dmd_cross_model_design.md`.